

Top FOSS Tools to Tech-Up the Teaching-Learning Experience

The teaching-learning process can be assisted, made more efficient and enhanced with the use of technology. Gen Y and Gen Z tend to learn better via their smartphones or the Internet. This article discusses five no-cost, open source teaching-learning tools that will greatly aid in the process. It is a good read for school administrators, teachers and students.

The entire teaching and learning process revolves around young minds. Getting and sustaining their attention and interest holds the key to success in this tricky process. Gen Y and Gen Z feel more comfortable using technology in their daily activities, thanks to the omnipresent smartphones and the Internet. Hence, educational institutions like schools, colleges and universities need to focus more on including technological tools to enhance the teaching-learning process to appeal to young minds. One of the myths that persists among many teachers and institution heads is regarding the costs associated with technology. They believe that these technological tools are not affordable in their environment. The aim of this article is to prove this myth wrong by introducing a few open source educational technology tools, which are available for almost no cost at all.

Technology will enhance various dimensions of the teaching-learning process, which spans a spectrum, as shown in Figure 1.

Sakai - the learning management system

Sakai is one of the leading open source learning management systems. It is widely used by many universities, colleges and schools across the globe to efficiently manage their academic activities. The Sakai Project was started in 2004 with an initial grant from the Mellon Foundation. To begin with, four universities were part of it. The largest contribution was from the University of Michigan in the US. The current release of Sakai is 10.3, and can be downloaded from <https://www.sakaiproject.org/node/105>. It is a state-of-the-art learning management system which techs-up the teaching, learning and research collaboration processes. The features of Sakai, shown in Figure 2, are as follows:

- Learning management with Sakai is fully loaded with various features like Assignments, Calendar, Chat, Discussion Forum, File Sharing, Grade Book, News, Profile, Syllabus, Test and Quizzes, Wiki, Lessons, etc. All of them are packaged with an intuitive user interface with an easy learning curve, an important factor while choosing an LMS.
- In addition to the learning features, Sakai has Research Collaboration. The Sakai Research Collaboration features include a rich set of tools for knowledge sharing, communication among peers in research, etc. As many universities encourage a research culture amongst its faculty and scholars, Sakai acts as a single-window solution for all such needs.
- Sakai is available in more than 20 languages, which is definitely a boon for non-English speaking communities. The supported languages include Arabic, Chinese, Japanese, Korean, Malay, Spanish, etc. Another advantage is that the language preferences can be set at either the system level, the course level or the project level. This allows instructors to run specific courses in particular languages.
- Inclusive education is an important factor that educators need to consider. Sakai has been created with accessibility

